MZUZU UNIVERSITY

Department of Mathematics

EMT 1101 - Precalculus

Supplementary Examinations

September 23, 2016 Time: 8 : 00 – 11 : 00 (3 hours)

1. Check that the paper has **10 questions.**
2. Attempt all questions
3. Write answers in the space provided
4. Show all your working clearly
5. Use proper notation
6. Simplify your answers as far as possible
7. **No Calculators**

1. (a) Simplify $\sqrt[3]{192} + \sqrt[3]{-81} + \sqrt{\sqrt[3]{9}}$. (3)

   (b) Factor $x^4 + x^2y^2 + y^4$ (Hint: Add and subtract a term). (3)

   (c) Simplify $\dfrac{x + \frac{xy}{y-x}}{\frac{y^2}{x^2-y^2} + 1}$. (3)

2. (a) Simplify $|x - 6| + |x - 7|$ given that $6 < x < 7$. (3)

   (b) Find all real solutions of the equation (3)

$$\frac{2}{x+4} - \frac{1}{x-4} = \frac{-7}{x^2-16}$$

3. A line passes through the points $(6, 2)$ and $(1, 4)$. Find the area of the triangle bounded by this line and the $x-$ and $y-$axes. (6)

4. Test the equation $y = 3^x - 3^{-x}$ for symmetry about:

   (a) the x-axis, (2)

   (b) the y-axis

   (c) the origin

5. (a) Determine the center and the radius of the circle whose equation is

$$9x^2 + 54x + 9y^2 - 6y + 64 = 0$$

   (b) Find the $x-$ and $y-$intercepts of the circle in part (a) where they exist. (5)

6. (a) Determine the value(s) of the constant k for which the equation $x^2 + 2(k+1)x + k^2 = 0$ has equal roots. (3)

   (b) Find all real solutions of each of the following equations:

   i. $y^{\frac{2}{3}} - 21y^{\frac{1}{3}} = -80$, (5)

   ii. $\dfrac{\sqrt{x} - 4}{\sqrt{x} - 18} = 3$. (4)

7. (a) Solve the following inequality leaving your answer in interval notation (5)

$$\left| \frac{x+1}{2} - \frac{(x-1)}{3} \right| < 1$$

   (b) The sum of the first n natural numbers is given by $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$. For which values of n will the sum be less than 210? (10)

8. (a) If $f(x) = mx + b$, show that $\frac{f(x_1)+f(x_2)}{2} = f\left(\frac{x_1+x_2}{2}\right)$. (3)

(b) Sketch the graph of (5)

$$g(x) = \begin{cases} x^3 & \text{when } x < 0 \\ \sqrt{x} & \text{when } 0 \leq x < 1 \\ \frac{1}{x} & \text{when } x \geq 1 \end{cases}$$

10. Let $f(x) = \dfrac{3}{x-1}$.

(a) Find the average rate of change of f on the interval $[4, 9]$. (2)

(b) Find f^{-1}. (3)

(c) Find the average rate of change of f^{-1} on the interval $[f(4), f(9)]$. What do you observe? (3)